End-Semester Exam - Data Structures (CS301)

Subject: Data Structures & Algorithms - CSE Sem 3

Centralized College Academic Repository

Part A [10 x 2 = 20 Marks]: Answer all questions.

1. Define Time Complexity and Big-O notation.
2. What is an AVL Tree and how does it balance?
3. Differentiate between BFS and DFS traversal.
4. Explain Infix to Postfix conversion rules.

Part B [5 x 16 = 80 Marks]: Comprehensive questions.

5. [a] Construct a Red-Black Tree for given keys.
5. [b] Explain Dijkstra’s shortest path algorithm with example.
6. Write quicksort algorithm and derive its average time complexity.